Zijian Guo

Contact Information

Center for Data Science
Zhejiang University
No. 866 Yuhangtang Road, Hangzhou, Zhejiang, China

zijguo@zju.edu.cn
https://zijguo.github.io

Education

Ph.D. Statistics, Wharton School, University of Pennsylvania — 2017
Thesis advisor: T. Tony Cai
"Statistical Inference For High-Dimensional Linear Models"

B.S. Mathematics, The Chinese University of Hong Kong — 2012
First Class Honor

Positions

Qiushi Chair Professor — Aug 2025- present
Center for Data Science
Zhejiang University

Associate Professor (with tenure) — July 2022- July 2025
Department of Statistics
Rutgers, the State University of New Jersey

Assistant Professor — Sep 2017- June 2022
Department of Statistics
Rutgers, the State University of New Jersey

Research Interests

My research focuses on reliable inference and generalizable learning under hidden confounding, high dimensionality, distribution shift, and nonregularity. My work spans causal inference, robust multi-source learning, nonstandard inference, and high-dimensional uncertainty quantification, with applications in genetics, proteomics, and multi-center health data.

Honors and Awards

- ICSA Outstanding Young Researcher Award — July. 2023
- Honorary Mention for Bernoulli Society New Researcher Award 2023 — Nov. 2022
- ICSA New Researcher Award, ICSA 2019 — Dec. 2019
- IMS travel Award, JSM — Aug. 2017
- President Gutmann Leadership Award, University of Pennsylvania — Apr. 2017
- J. Parker Bursk Prize — Sept. 2016
  Awarded by the Wharton School for excellence in research.
- Statistics in Epidemiology Young Investigator Award, JSM — Aug. 2013
  Awarded by the ASA section on Statistics in Epidemiology.

Visiting Positions

Invited Research Fellow (Nov 2018, Sep-Oct 2023, Apr-May 2024, June 2025)
Forschungsinstitut für Mathematik, ETH, Zürich (Host: Peter Bühlmann)

Invited Research Fellow (Apr-May 2024, Dec 2024, June 2025)
Inria, Paris (Host: Francis Bach)

Invited Research Fellow (June 2024)
Department of Statistics/Economics, University of Oxford (Host: Frank Windmeijer)

Invited Research Fellow (Apr 2024)
Department of Mathematics, Cambridge University (Host: Richard Samworth)

Invited Research Fellow (March 2024)
University of Twente (Host: Johannes Schmidt-Hieber)

Invited Research Fellow (Sep 2019, Sep 2023, Feb-Mar 2024)
The Chan School of Public Health, Harvard University (Host: Tianxi Cai)

Invited Research Fellow (Feb 2024)
Department of Economics, Yale University (Host: Xiaohong Chen)

Invited Research Fellow (Jan-Feb 2024)
Statistics & Data Science, Carnegie Mellon University (Host: Larry Wasserman)

Invited Research Fellow (Aug 2017)
Perelman School of Medicine, University of Pennsylvania (Host: Hongzhe Li)

Representative work

* indicates alphabetical ordering authorship; underline indicates supervised students; # indicates equal contribution; ✉ indicates corresponding authorship.

(R1) *Cai, T. T., and **Guo, Z.** (2017). Confidence intervals for high-dimensional linear regression: Minimax rates and adaptivity. *Annals of Statistics*, 45(2), 615-646.

(R2) **Guo, Z.**, Kang, H., Cai, T. T., and Small, D. S. (2018). Confidence Intervals for Causal Effects with Invalid Instruments using Two-Stage Hard Thresholding with Voting. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 80(4), 793-815.

(R3) **Guo, Z.**#✉, Ćevid, D.#, and Bühlmann, P. (2022). Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding. *Annals of Statistics*, 50(3), 1320-1347.

(R4) **Guo, Z.** (2023). Causal Inference with Invalid Instruments: Post-selection Problems and A Solution Using Searching and Sampling. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 85(3), 959-985.

(R5) **Guo, Z.** (2024). Statistical Inference for Maximin Effects: Identifying Stable Associations across Multiple Studies. *Journal of the American Statistical Association*, 119(547), 1968-1984.

(R6) Yao, M., Miller, G., Vardarajan, B., Baccarelli, A., **Guo, Z.** ✉, and Liu, Z. ✉. (2024). Deciphering Proteins in Alzheimer's Disease: A New Mendelian Randomization Method Integrated with AlphaFold3 for 3D Structure Prediction. *Cell Genomics*, 4(12), 100700.

(R7) Wang, Z.#, Hu, Y.#, Bühlmann, P.✉, & **Guo, Z.**✉(2024). Causal Invariance Learning via Efficient Nonconvex Optimization. *arXiv preprint arXiv:2412.11850.*

(R8) **Guo, Z.**, Wang, Z., Hu, Y., and Bach, F. (2025). Statistical Analysis of Conditional Group Distributionally Robust Optimization with Cross-Entropy Loss. *arXiv preprint arXiv:2507.09905.*

(R9) Zheng, M., Bonvini, M.✉, and **Guo, Z.**✉(2025). Perturbed Double Machine Learning: Nonstandard Inference Beyond the Parametric Length. *arXiv preprint arXiv:2511.01222*

Preprints

* indicates alphabetical ordering authorship; underline indicates supervised students; # indicates equal contribution; ✉ indicates corresponding authorship

1. Shi, Y., Liu, M.✉, and **Guo, Z.**✉(2026). TabCUQ: Context-Uncertainty Quantification for Tabular Foundation Models. *arXiv preprint arXiv:2610.39558.*
2. Shen, X., **Guo, Z.**, and Bach, F. (2026). Generalized Engression Models. *arXiv preprint arXiv:2610.01823.*

3. Xiong, X., **Guo, Z.**, and Cai, T. (2026). Guided Uncertainty-Aware Robust Domain Transfer. *arXiv preprint arXiv:2609.35654.*

4. Chen, J., **Guo, Z.**#✉, and Shen, X.#✉(2026). Error Analysis of Neural-Network-Based Engression. *arXiv preprint arXiv:2607.27723.*

5. Mei, Z., Fan, Q., and **Guo, Z.** (2026). Identification and Robust Inference for Multiple Treatments with Possibly Invalid Instruments. *arXiv preprint arXiv:2607.21481.*

6. Xu, S., **Guo, Z.**, Staveland, B. R., Knight, R. T., and Li, L.✉(2026). Robust Learning of Heterogeneous Dynamic Systems. *arXiv preprint arXiv:2604.05285.*

7. Heng, S.#✉, Shen, Y.#, **Guo, Z.**✉(2026). Propensity Score Propagation: A General Framework for Design-Based Inference with Unknown Propensity Scores. *arXiv preprint arXiv:2601.13150*

8. Xiong, X., **Guo, Z.**, Zhu, H., Hong, C., Smoller, J. W., Cai, T., Liu, M.✉(2026). Adversarial Drift-Aware Predictive Transfer: Toward Durable Clinical AI. *arXiv preprint arXiv:2601.11860*

9. Koo, T. and **Guo, Z.**✉(2025). Synthetic Control with Weight Uncertainty: Robust Identification and Statistical Inference. *arXiv preprint arXiv:2511.02632*

10. Zheng, M., Bonvini, M.✉, and **Guo, Z.**✉(2025). Perturbed Double Machine Learning: Nonstandard Inference Beyond the Parametric Length. *arXiv preprint arXiv:2511.01222*

11. **Guo, Z.**, Wang, Z., Hu, Y., and Bach, F. (2025). Statistical Analysis of Conditional Group Distributionally Robust Optimization with Cross-Entropy Loss. *arXiv preprint arXiv:2507.09905.*

12. Wang, Z., Liu, M., Lei, J., Bach, F.✉, & **Guo, Z.**✉(2025). StablePCA: Distributionally Robust Learning of Shared Representations from Multi-Source Data. *arXiv preprint arXiv:2505.00940.*

13. Gu, Y., Fang, C., Xu, Y., **Guo, Z.**, & Fan, J. (2025). Fundamental Computational Limits in Pursuing Invariant Causal Prediction and Invariance-Guided Regularization. *arXiv preprint arXiv:2501.17354.*

14. Wang, Z.#, Hu, Y.#, Bühlmann, P.✉, & **Guo, Z.**✉(2024). Causal Invariance Learning via Efficient Nonconvex Optimization. *arXiv preprint arXiv:2412.11850.*

15. Rakshit, P., & **Guo, Z.** (2024). Statistical Inference in High-dimensional Poisson Regression with Applications to Mediation Analysis. *arXiv preprint arXiv:2410.20671.*

16. Wang, Z., Si, N., **Guo, Z.**, and Liu, M.✉(2024). Inference for Stable Variable Importance across Multi-source Domains via Adversarial Learning.
arXiv preprint arXiv:2409.07380.

17. Zhan, K., Xiong, X., **Guo, Z.**, Cai, Tianxi, and Liu, M. (2024). Domain Adaptation Optimized for Robustness in Mixture Populations. *arXiv preprint arXiv:2407.20073.*

18. *Fan, Q., **Guo, Z.**, Mei, Z., and Zhang, C. (2023). Inference for Nonlinear Endogenous Treatment Effects Accounting for High-Dimensional Covariate Complexity. *arXiv preprint arXiv:2310.08063.*

19. Zhu, J., Zhang, J., **Guo, Z.**, and Heng, S.✉(2023). Randomization-Based Inference for Average Treatment Effects in Inexactly Matched Observational Studies. *arXiv preprint arXiv:2308.02005.*

PUBLICATIONS

39 peer-reviewed publications. * indicates alphabetical ordering authorship; underline indicates supervised students; # indicates equal contribution; ✉ indicates corresponding authorship.

1. Xiong, X., **Guo, Z.**✉, and Cai, T.✉(2026). Guided Adversarial Robust Transfer Learning with Source Mixing. *Journal of the American Statistical Association*, 1–14. Published online September 8, 2026.
2. Liu, Y.#, Liu, M.#✉, **Guo, Z.**, and Cai, Tianxi. (2026). Efficient Modeling of Surrogates to Improve Multi-source High-dimensional Integrative Regression. *Journal of Machine Learning Research*, 27(143), 1–53.
3. Wang, Z., Bühlmann, P., and **Guo, Z.** (2026). Distributionally Robust Learning for Multisource Unsupervised Domain Adaptation. *Annals of Statistics*, 54(2), 570–596.
4. **Guo, Z.**, Zheng, M., and Bühlmann, P. (2026). Robustness Against Weak or Invalid Instruments: Exploring Nonlinear Treatment Models with Machine Learning. *Journal of Machine Learning Research*, 27(198), 1–67.
5. ***Guo, Z.**, Yuan, W. and Zhang, C. (2026). Decorrelated Local Linear Estimator: Inference for Non-linear Effects in High-dimensional Additive Models. *Journal of Machine Learning Research*, 27(27), 1–79.
6. Chang, T. H., **Guo, Z.**, and Malinsky, D. (2026). Post-selection inference for causal effects after causal discovery. *Biometrika*, 113(1), asaf073.
7. Scheidegger, C., **Guo, Z.**, & Bühlmann, P. (2026). Inference for Heterogeneous Treatment Effects with Efficient Instruments and Machine Learning. *Electronic Journal of Statistics*, 20(1), 718–770.
8. **Guo, Z.**#✉, Li, X.#, Han, L., and Cai, Tianxi. (2025). Robust Inference for Federated Meta-Learning. *Journal of the American Statistical Association*, 120(551), 1695–1710.
9. Scheidegger, C., **Guo, Z.**, and Bühlmann, P. (2025). Spectral Deconfounding for High-Dimensional Sparse Additive Models. *ACM/IMS Journal of Data Science*, 2(1), Article 1, 52 pages.
10. Lin, Y., **Guo, Z.**, Sun, B., and Lin, Z. (2025). Testing High-Dimensional Mediation Effect with Arbitrary Exposure-Mediator Coefficients. *Test*, 34(3), 530–579.
11. Carl, D., Emmenegger, C., Bühlmann, P., and **Guo, Z.** (2025). TSCI: Two Stage Curvature Identification for Causal Inference with Invalid Instruments in R. *Journal of Statistical Software*, 114(7), 1–21.
12. *Fan, Q., **Guo, Z.**, and Mei, Z. (2025). A Heteroscedasticity-Robust Overidentifying Restriction Test with High-Dimensional Covariates. *Journal of Business & Economic Statistics*, 43(2), 413–422.
13. Kang, H., **Guo, Z.**, Liu, Z., and Small, D. (2025). Identification and Inference with Invalid Instruments. *Annual Review of Statistics and Its Application*, 12, 385–405.
14. **Guo, Z.** (2024). Statistical Inference for Maximin Effects: Identifying Stable Associations across Multiple Studies. *Journal of the American Statistical Association*, 119(547), 1968-1984.
15. Yao, M., Miller, G., Vardarajan, B., Baccarelli, A., **Guo, Z.** ✉, and Liu, Z. ✉. (2024). Deciphering Proteins in Alzheimer's Disease: A New Mendelian Randomization Method Integrated with AlphaFold3 for 3D Structure Prediction. *Cell Genomics*, 4(12), 100700.
16. Ma, R., **Guo, Z.**, Cai, T. T., and Li, H.✉(2024). Statistical Inference for Genetic Relatedness Based on High-Dimensional Logistic Regression. *Statistica Sinica*, 34(2), 1023–1043.

17. Rakshit, P., Wang, Z., Cai, T. T., and **Guo, Z.** (2024). `SIHR`: Statistical Inference in High-Dimensional Linear and Logistic Regression Models. *The R Journal*, 16(3), 27–45.

18. *Cai, T. T., **Guo, Z.**, and Xia, Y. (2023). Statistical inference and large-scale multiple testing for high-dimensional regression models. *Test (with discussion)*, 32(4), 1135-1171.

19. **Guo, Z.** (2023). Causal Inference with Invalid Instruments: Post-selection Problems and A Solution Using Searching and Sampling. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 85(3), 959-985.

20. *Cai, T. T., **Guo, Z.**, and Ma, R. (2023). Statistical Inference for High-Dimensional Generalized Linear Models with Binary Outcomes. *Journal of the American Statistical Association*, 118 (542), 1319-1332.

21. Koo, T., Lee, Y., Small, D. S., and **Guo, Z.** (2023). `RobustIV` and `controlfunctionIV`: Causal Inference for Linear and Nonlinear Models with Invalid Instrumental Variables. *Observational Studies*, 9(4), 97-120.

22. Hou, J., **Guo, Z.**, and Cai, T. (2023). Surrogate assisted semi-supervised inference for high dimensional risk prediction. *Journal of Machine Learning Research*, 24(265), 1-58.

23. Wang, X., Zhou, H., ⋯, 4CE, Avillach, P.✉, **Guo, Z.**✉, and Cai, Tianxi✉. (2022). Surv-Maximin: Robust Federated Approach to Transporting Survival Risk Prediction Models. *Journal of Biomedical Informatics*, 134 (2022): 104176.

24. **Guo, Z.**#✉, Ćevid, D.#, and Bühlmann, P. (2022). Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding. *Annals of Statistics*, 50 (3), 1320 - 1347.

25. ***Guo, Z.** and Zhang, C. (2022). Extreme Eigenvalues of Nonlinear Correlation Matrices with Applications to Additive Models. *Stochastic Processes and Their Applications*, 150, 1037-1058.

26. **Guo, Z.**, Renaux, C., Bühlmann, P., and Cai, T. T. (2021). Group Inference in High Dimensions with Applications to Hierarchical Testing. *Electronic Journal of Statistics*, 15(2), 6633-6676.

27. **Guo, Z.**, Rakshit, P., Herman, D., and Chen, J. (2021). Inference for the Case Probability in High-dimensional Logistic Regression. *Journal of Machine Learning Research*, 22(254), 1-54.

28. *Cai, Tianxi, Cai, T. T., and **Guo, Z.** (2021). Optimal Statistical Inference for Individualized Treatment Effects in High-dimensional Models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 83(4), 669-719.

29. *Cai, T. T., and **Guo, Z.** (2020). Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 82(2), 391-419.

30. **Guo, Z.**, Wang, W., Cai, T. T., and Li, H. (2019). Optimal Estimation of Genetic Relatedness in High-dimensional Linear Models. *Journal of the American Statistical Association*, 114(525), 358-369.

31. **Guo, Z.**, Kang, H., Cai, T. T., and Small, D. S. (2018). Testing Endogeneity with High Dimensional Covariates. *The Journal of Econometrics*, 207(1), 175-187.

32. **Guo, Z.**, Kang, H., Cai, T. T., and Small, D. S. (2018). Confidence Intervals for Causal Effects with Invalid Instruments using Two-Stage Hard Thresholding with Voting. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 80(4), 793-815.

33. *Cai, T. T., and **Guo, Z.** (2018). Accuracy assessment for high-dimensional linear regression. *Annals of Statistics*, 46(4), 1807-1836.

34. **Guo, Z.**, Small, D. S., Gansky, S. A., and Cheng, J. (2018). Mediation analysis for count and zero-inflated count data without sequential ignorability and its application in dental studies. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 67(2), 371-394.
35. Cheng, J., Cheng N. F., **Guo, Z.**, Gregorich, S., Ismail, A. I., and Gansky, S. A. (2018). Mediation analysis for count and zero-inflated count data. *Statistical Methods in Medical Research*, 27(9), 2756-2774.
36. *Cai, T. T., and **Guo, Z.** (2017). Confidence intervals for high-dimensional linear regression: Minimax rates and adaptivity. *Annals of Statistics*, 45(2), 615-646.
37. **Guo, Z.**, and Small, D. S. (2016). Control function instrumental variable estimation of nonlinear causal effect models. *Journal of Machine Learning Research*, 17(100), 1-35.
38. **Guo, Z.**, Cheng, J., Lorch, S. A., and Small, D. S. (2014). Using an instrumental variable to test for unmeasured confounding. *Statistics in Medicine*, 33(20), 3528-3546.
39. **Guo, Z.**, Kogan, R., Qiu, H., and Strichartz, R. S. (2014). Boundary value problems for a family of domains in the Sierpinski gasket. *Illinois Journal of Mathematics*, 58(2), 497-519.

Software

1. Python package **engression**: generalized engression models for distributional regression with continuous, discrete, and mixed-type outcomes. Available at `https://github.com/xwshen51/engression`
2. R package **SIHR**: statistical inference in high-dimensional regression. Available at `https://cran.r-project.org/web/packages/SIHR/index.html`
3. R package **RobustIV**: robust causal inference with possibly invalid instruments. Available at `https://cran.r-project.org/web/packages/RobustIV/index.html`
4. R package **TSCI**: two stage curvature identification with machine learning. Available at `https://cran.r-project.org/web/packages/TSCI/index.html`
5. R package **maczic**: mediation analysis for count and zero-inflated count data. Available at `https://cran.r-project.org/web/packages/maczic/index.html`.
6. R package **MaximinInfer**: inference for maximin effects in high-dimensional settings. Available at `https://cran.r-project.org/web/packages/MaximinInfer/index.html`
7. R package **DLL**: inference for function derivative in high-dimensional additive models. Available at `https://cran.r-project.org/web/packages/DLL/index.html`
8. R package **DDL**: inference for regression parameters in high-dimensional models with hidden confounders. Available at `https://cran.r-project.org/web/packages/DDL/index.html`
9. R package **controlfunctionIV**: control function method with possibly invalid instruments. Available at `https://cran.r-project.org/web/packages/controlfunctionIV/index.html`

R codes are available at `https://github.com/zijguo`.

Teaching Experience

Course Instructor

- **Zhejiang University**

  Optimization for Data Science 2026

- **Rutgers University (Ph.D. level)**

  STAT 593: Theory of Statistics Fall 2024

STAT 593: Theory of Statistics — Fall 2023

STAT 594: Advanced Modern Statistical Inference II — Spring 2019

Instructor Rating: 4.82 out of 5.0

- **Rutgers University (Master level)**

FSRM 588: Financial Data Mining — Spring 2025

FSRM 588: Financial Data Mining — Fall 2021

Instructor Rating: 4.75 out of 5.0

FSRM 588: Financial Data Mining (Virtual) — Spring 2021

FSRM 588: Financial Data Mining — Spring 2020

Instructor Rating: 4.50 out of 5.0

FSRM 588: Financial Data Mining — Fall 2019

Instructor Rating: 4.75 out of 5.0

FSRM 588: Financial Data Mining — Fall 2018

Instructor Rating: 4.71 out of 5.0

FSRM 588: Financial Data Mining — Fall 2017

Instructor Rating: 4.82 out of 5.0

- **Rutgers University (Undergraduate level)**

STAT 384: Intermediate Statistical Analysis (Virtual) — Spring 2021

STAT 484: Basic Applied Statistics (Virtual)

- **The Wharton School, University of Pennsylvania**

STAT 111 : Introductory Statistics — Summer 2016

Recitation Instructor — Fall 2014

The Wharton School, University of Pennsylvania

STAT 111: Introductory Statistics

Teaching Assistant

The Wharton School, University of Pennsylvania

STAT 102: Business Statistics — Spring 2017

STAT 970: Mathematical Statistics — Fall 2016

STAT 622: Statistical Modeling — Spring 2016

STAT 550: Mathematical Statistics — Fall 2015

Student Supervision

PhD Thesis Advisor:

1. Prabrisha Rakshit (Graduated in 2023; Assistant Professor in the Department of Operations Management, Quantitative Methods, and Information Systems at the Indian Institute of Management Udaipur);
2. Taehyeon Koo (Graduated in 2025; Joint with Dr. Nicole Pashley; first position: Postdoctoral fellow at Columbia University);
3. Zhenyu Wang (Rutgers, Expected 2027);
4. Mengchu Zheng (Rutgers, Expected 2028);
5. Chao Qin (Zhejiang University, Expected 2028)
6. Yuebei Shi (Zhejiang University, Expected 2028)
7. Chen Ling (Zhejiang University, Expected 2030)
8. Jiahang Shao (Zhejiang University, Expected 2030)

9. Wenyuan Shu (Zhejiang University, Expected 2031)

Postdoc Advisor:

1. Yichuan Bai (Ph.D. at Iowa State)

Undergraduate Research Assistant: Yuyang Tan (Zhejiang University)

Academic Service

1. Chair, Organizing Committee, Frontiers in Data Science and Intelligent Decision-Making, Zhejiang University, November 13, 2026 (scheduled)
2. Chair, Local Organizing Committee, 2026 International Conference on Frontiers of Data Science, Zhejiang University
3. Associate Editor
   - *Journal of the American Statistical Association*, 2023–present
   - *TEST*
   - *Electronic Journal of Statistics*
4. Organizer of Causal Inference Reading Group (Joint with Nicole Pashley and Tirthankar Dasgupta), Department of Statistics, Rutgers, 2017–2025
5. Department Retreat Chair (2019-2020), Department of Statistics, Rutgers
6. Department Seminar Chair (2018-2019), Department of Statistics, Rutgers
7. Other Rutgers Committee service:
   - Department website committee (2020-2023)
   - Department retreat committee (2017-2018)
   - FSRM/MSDS committee (2017-2023)
   - Ph.D. Exam committee (2018-2020)
   - Graduate Curriculum committee (2019-2021)
8. Organizing Committee for 2019 Rutgers Statistics Symposium
9. Program Committee for ICSA 2019 11th International Conference
10. Local Organizing Committee for 2018 ICSA Applied Symposium.
11. Reviewer for the following journals: *Annals of Statistics*, *Journal of the American Statistical Association*, *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, *Biometrika*, *Journal of Machine Learning Research*, *Journal of Econometrics*, *Biometrics*,*Statistica Sinica*, *IEEE International Symposium on Information Theory*, *Journal of Applied Statistics*, *COLT*.

Short Courses

1. *"Nonregular Inference"*, Department of Economics, Yale University, Feb. 2024
2. *"Nonregular Inference"*, CELEHS Distinguished Lecture Series, Department of Biostatistics, Harvard University, Feb. 2024
3. *"Robust Deconfounding"*, Department of Biostatistics, Harvard University, Feb. 2024

Invited Talks

1. Statistics and Data Science Seminar, Center for Statistical Science, Peking University, Beijing, China, *"Nonregular Inference."*, September 29, 2026
2. Mathematics Colloquium, Department of Mathematics, HKUST, Hong Kong, China, *"Nonregular Causal Inference."*, March 2026
3. S. S. Wilks Memorial Seminar in Statistics, Department of Operations Research and Financial Engineering, Princeton University, USA, *"Learning Shared Representations with Multi-Source Data: A Distributionally Robust Optimization Approach."*, Feb. 2025
4. Department seminar, Department of Economics, University of Oxford, UK, *"Adversarially Robust Learning: Identification, Estimation, and Uncertainty Quantification."*, June 2024
5. Department seminar, Department of Statistics, University of Oxford, UK, *"Robust Instrumental Variable Analysis."*, June 2024
6. Department seminar, University of Geneva, Switzerland, *"Adversarially Robust Learning: Identification, Estimation, and Uncertainty Quantification."*, May 2024
7. Department seminar, EPFL, Switzerland, *"Adversarially Robust Learning: Identification, Estimation, and Uncertainty Quantification."*, May 2024
8. Department seminar, Seminar for Statistics, ETH Zurich, Switzerland, *"Adversarially Robust Learning: Identification, Estimation, and Uncertainty Quantification."*, May 2024
9. Department seminar, Inria, France, *"Adversarially Robust Learning: Identification, Estimation, and Uncertainty Quantification."*, April 2024
10. Seminar, Amsterdam Machine Learning Lab, *"Adversarially Robust Learning: Identification, Estimation, and Uncertainty Quantification."*, March 2024
11. Department seminar, Department of Economics, Yale University, USA, *"Robust Instrumental Variable Analysis."*, Feb 2024
12. Causal Reading Group, Department of Statistics & Data Science, Carnegie Mellon University, USA, *"Robust Deconfounding."*, Jan 2024
13. Machine Learning Reading Group, Department of Statistics & Data Science, Carnegie Mellon University, USA, *"Adversarially Robust Learning: Identification, Estimation, and Uncertainty Quantification."*, Jan 2024
14. Department seminar, Department of Statistics, The Hong Kong Polytechnic University, Hong Kong, *"Domain Generalization with Adversarially Robust Learning: Identification, Estimation, and Uncertainty Quantification."*, Jan 2024
15. Department seminar, Department of Statistics, The Chinese University of Hong Kong, Hong Kong, *"Domain Generalization with Adversarially Robust Learning: Identification, Estimation, and Uncertainty Quantification."*, Jan 2024
16. Invited talk, ICSDS, Lisbon, Portugal, *"Distributionally Robust Machine Learning with Multi-source Data."*, Sep 2023
17. Department seminar, Department of Mathematics, NJIT, USA, *"Two Stage Curvature Identification with Machine Learning: Causal Inference with Possibly Invalid Instrumental Variables."*, Oct 2023
18. ETH/UZH Research Seminar on Statistics and FDS seminar series, Zurich, Switzerland, *"Robust Causal Inference with Possibly Invalid Instruments: Post-selection Problems and A Solution Using Searching and Sampling."*, Sep 2023
19. Harvard Cai Lab seminar, The Chan School of Public Health, Harvard University, USA, *"Distributionally Robust Machine Learning with Multi-source Data."*, Sep 2023
20. Department seminar, ISOM, HKUST, Hong Kong, China, *"Robust Inference for Federated Meta-Learning."*, Aug 2023
21. Invited talk, 2023 Hangzhou Data Science Conference, Hangzhou, China, *"Statistical Inference for Maximin Effects: Identifying Stable Associations across Multiple Studies."*, Aug 2023
22. Invited speaker (virtual), EcoSta 2023, Tokyo, Japan, *"Two Stage Curvature Identification with Machine Learning: Causal Inference with Possibly Invalid Instrumental*

Variables.", Aug 2023

23. Invited speaker, 9th RUC-IFS, Beijing, China, *"Statistical Inference for Maximin Effects: Identifying Stable Associations across Multiple Studies."*, July 2023
24. Invited speaker, JCSDS, Beijing, China, *"Robust Inference for Federated Meta-Learning."*, July 2023
25. Invited speaker, ICSA 2023, Hong Kong, China, *"Causal Inference with Invalid Instruments: Post-selection Problems and A Solution Using Searching and Sampling."*, July 2023
26. Invited talk (virtual), joint statistics seminar of ENSAE/Ecole Polytechnique, France, *"Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding."*, June 2023
27. Invited talk (virtual), Inria, France, *"Statistical Inference for Maximin Effects: Identifying Stable Associations across Multiple Studies."*, May 2023
28. Invited talk (virtual), Beijing Normal University, China, *"Statistical Inference for Maximin Effects: Identifying Stable Associations across Multiple Studies."*, May 2023
29. Invited talk (virtual), Peking University, China, *"Robust Inference for Federated Meta-Learning."*, March 2023
30. Invited talk (virtual), 1st joint webinar of the IMS New Researchers Group, Young Data Science Researcher Seminar Zürich and the YoungStatS Project, *"Statistical Inference for Maximin Effects: Identifying Stable Associations across Multiple Studies."*, March 2023
31. Invited talk (virtual), CMStatistics 2022, London, UK, *"Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding."*, Dec 2022
32. Levin Lecture, Department of Biostatistics, Columbia University, USA, *"Multi-data Replicability via the Maximin Criterion: High-Dimensional Inference for Group Distributionally Robust Models."*, Sep 2022
33. Lab seminar (virtual) led by Tianxi Cai, The Chan School of Public Health, Harvard University, USA, *"Robust Inference for Federated Meta-Learning."*, Sep 2022
34. RAND Center Causal Inference Symposium (virtual), *"Two Stage Curvature Identification with Machine Learning: Causal Inference with Possibly Invalid Instrumental Variables."*, Aug 2022
35. Invited speaker, JSM 2022, USA, *"Causal Inference with Invalid Instruments: Post-selection Problems and A Solution Using Searching and Sampling."*, Aug 2022
36. Invited speaker (virtual), ICSA 2022, University of Florida, USA, *"Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding."*, June 2022
37. Online causal inference seminar (virtual, discussed by Professor Frank Windmeijer), *"Two Stage Curvature Identification with Machine Learning: Causal Inference with Possibly Invalid Instrumental Variables."*, April 2022
38. Center for Causal Inference seminar (virtual), University of Pennsylvania, USA, *"Two Stage Curvature Identification with Machine Learning: Causal Inference with Possibly Invalid Instrumental Variables."*, March 2022
39. Department seminar (virtual), Department of Biostatistics & Medical Informatics, University of Wisconsin–Madison, USA, *"Decorrelated Local Linear Estimator: Inference for Non-linear Effects in High-dimensional Additive Models."*, Feb 2022
40. Lab seminar (virtual) led by Tianxi Cai, The Chan School of Public Health, Harvard University, USA, *"Transfer Learning with Multi-source Data: High-Dimensional Inference for Group Distributionally Robust Models."*, Jan 2022
41. Invited speaker (virtual), CMStatistics 2021, *"Inference for Case Probability in High-dimensional Logistic Regression."*, Dec 2021
42. Invited speaker (virtual), JSM 2021, Seattle, USA, *"Inference for High-dimensional Maximin Effects in Heterogeneous Regression Models Using a Sampling Approach."*, Aug 2021
43. Invited speaker (virtual), The First International Conference on Statistics and Related

Fields, Luxembourg, *"Inference for High-dimensional Maximin Effects in Heterogeneous Regression Models Using a Sampling Approach."*, July 2021

44. Lab seminar (virtual) led by Tianxi Cai, The Chan School of Public Health, Harvard University, USA, *"Maximin Effect and Distributional Robustness: A Review and New Advances."*, June 2021
45. Department seminar (virtual), Department of Statistics, East China Normal University, Shanghai, China, *"Inference for High-dimensional Maximin Effects in Heterogeneous Regression Models Using a Sampling Approach."*, May 2021
46. Center for Causal Inference seminar (virtual), University of Pennsylvania, USA, *"Post-selection Problems for Causal Inference with Invalid Instruments: A Solution Using Searching and Sampling."*, May 2021
47. Department seminar (virtual), Department of Economics, The Chinese University of Hong Kong, Hong Kong, China, *"Post-selection Problems for Causal Inference with Invalid Instruments: A Solution Using Searching and Sampling."*, April 2021
48. Department seminar (virtual), Department of Statistics, The University of Hong Kong, Hong Kong, China, *"Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding."*, April 2021
49. Department seminar (virtual), Medical School, University of Exeter, Exeter, UK, *"Inference for Non-linear Treatment Effects with Control Function Methods."*, Feb 2021
50. Invited talk (virtual), CMStatistics 2020, London, UK, *"Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding."*, Dec 2020
51. Department seminar (virtual), Department of Statistics, Cornell University, USA, *"Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding."*, Oct 2020
52. Invited talk (virtual), JSM 2020, Philadelphia, USA, *"Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding and Measurement Errors."*, Aug 2020
53. Invited participant (virtual), *"Mathematical and Statistical Challenges in Uncertainty Quantification."*, Cambridge, UK, July 2020
54. Department seminar (virtual), Department of Statistics, UC Davis, USA, *"Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding and Measurement Errors."*, May 2020
55. Center for Causal Inference seminar (virtual), University of Pennsylvania, USA, *"Doubly Debiased Lasso: High-Dimensional Inference under Hidden Confounding and Measurement Errors."*, May 2020
56. Biostatistics reading group (virtual, led by Jinbo Chen), University of Pennsylvania, USA, *"Group Inference in High Dimensions with Applications to Hierarchical Testing."*, May 2020
57. Department seminar, Department of Statistics, East China Normal University, Shanghai, China, *"Group Inference in High Dimensions with Applications to Hierarchical Testing."*, Dec 2019
58. Invited talk, 11th ICSA International Conference, Hangzhou, China, *"Group Inference in High Dimensions with Applications to Hierarchical Testing."*, Dec 2019
59. Invited talk, International Statistical Conference in Memory of Professor Sik-Yum Lee, Hong Kong, China, *"Group Inference in High Dimensions with Applications to Hierarchical Testing."*, Dec 2019
60. Causal reading group (led by James Robins), School of Public Health, Harvard University, USA, *"Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications."*, Sep 2019
61. Department seminar, Department of Statistics, East China Normal University, Shanghai, China, *"Individualized Treatment Selection: A Hypothesis Testing Approach in High-dimensional Models."*, June 2019
62. Invited talk, 2019 Hangzhou Data Science Conference, Hangzhou, China, *"Local Inference in High-dimensional Sparse Additive Modeling."*, May 2019
63. Department seminar, School of Data Science, City University of Hong Kong, Hong

Kong, China, *"Individualized Treatment Selection: A Hypothesis Testing Approach in High-dimensional Models."*, May 2019

64. Department seminar, ISOM, HKUST, Hong Kong, China, *"Local Inference in High-dimensional Sparse Additive Modeling."*, May 2019
65. Department seminar, Department of Statistics, University of Virginia, USA, *"Local Inference in High-dimensional Sparse Additive Modeling."*, March 2019
66. Invited talk, 2019 ICSA Data Science Conference, Xishuangbanna, Yunnan, China, *"Individualized Treatment Selection: A Hypothesis Testing Approach in High-dimensional Models."*, Jan 2019
67. Young Research Session, Memorial Workshop for Lawrence D. Brown, University of Pennsylvania, USA, *"Individualized Treatment Selection: A Hypothesis Testing Approach in High-dimensional Models."*, Nov 2018
68. Department seminar, Seminar for Statistics, Department of Mathematics, ETH Zurich, Switzerland, *"Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications."*, Nov 2018
69. Department seminar, Department of Mathematics, NJIT, USA, *"Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications."*, Nov 2018
70. Department seminar, ORFE, Princeton University, USA, *"Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications."*, Oct 2018
71. Department seminar, ISOM, HKUST, Hong Kong, China, *"Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications."*, July 2018
72. Department seminar, Department of Statistics, Nankai University, China, *"Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications."*, July 2018
73. Invited talk, IMS Asia Pacific Rim Meeting, Singapore, *"Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications."*, June 2018
74. Invited talk, Hong Kong EcoStat Conference, Hong Kong, China, *"Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications."*, June 2018
75. Invited talk, ICSA Symposium 2018, New Brunswick, USA, *"Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications."*, June 2018
76. Invited talk, Purdue Symposium on Statistics, USA, *"Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications."*, June 2018
77. Invited talk, 2018 Hangzhou Data Science Conference, Hangzhou, China, *"Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications."*, May 2018
78. Invited talk, Lorentz Center, Leiden University, Netherlands, *"Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications."*, Apr 2018
79. Department seminar, Department of Statistics, Columbia University, USA, *"Semi-supervised Inference for Explained Variance in High-dimensional Linear Regression and Its Applications."*, Apr 2018
80. Invited talk, *"Statistical Foundations of Uncertainty Quantification for Inverse Problems."*, Cambridge, UK, June 2017
81. Seminar, Center for Statistical Methods in Big Data, University of Pennsylvania, USA, *"Inference with High-dimensional Covariates and Possibly Invalid Instruments."*, Apr 2017

82. Seminar, Institute of Data Science, Fox School of Business, Temple University, USA, *"Inference for High-dimensional Linear Models: Fundamental Limits and Algorithms."*, Feb 2017
83. Department seminar, Department of Biostatistics, UC Berkeley, USA, *"Inference for High-dimensional Linear Models: Fundamental Limits and Algorithms."*, Feb 2017
84. Department seminar, Department of Statistics, Rutgers University, USA, *"Inference for High-dimensional Linear Models: Fundamental Limits and Algorithms."*, Feb 2017
85. Department seminar, Department of Statistics, University of Michigan, USA, *"Inference for High-dimensional Linear Models: Fundamental Limits and Algorithms."*, Jan 2017
86. Department seminar, Department of Statistics, University of Minnesota, USA, *"Inference for High-dimensional Linear Models: Fundamental Limits and Algorithms."*, Jan 2017
87. Department seminar, Department of Statistics, UIUC, USA, *"Inference for High-dimensional Linear Models: Fundamental Limits and Algorithms."*, Jan 2017
88. Department seminar, DPMMS, University of Cambridge, UK, *"Inference for High-dimensional Linear Regression: Fundamental Limits and Algorithms."*, Jan 2017
89. Department seminar, Department of Statistics, UC Santa Barbara, USA, *"Inference for High-dimensional Linear Models: Fundamental Limits and Algorithms."*, Jan 2017
90. Invited talk, Mathematical Meeting in Statistics, Fréjus, France, *"Optimal Estimation of Genetic Correlation in High-dimensional Linear Models."*, Dec 2016
91. Econometrics Lunch, Department of Economics, University of Pennsylvania, USA, *"Confidence Intervals for Treatment Effects in High-dimensional Linear Models."*, Nov 2016

Other Talks

1. Topic contributed talk, Joint Statistical Meetings, Baltimore, USA, *"Optimal Estimation of Co-Heritability in High-Dimensional Linear Models"*, Aug. 2017
2. Contributed talk, Joint Statistical Meetings, Chicago, USA, *"Accuracy Assessment for High-dimensional Linear Regression"*, Aug. 2016
3. Contributed talk, Eastern North American Region, Austin, USA, *"Confidence Intervals for High-Dimensional Linear Regression: Minimax Rates and Adaptivity"*, Mar. 2016
4. Poster presentation, John W. Tukey Conference, Princeton University, *"Confidence Intervals for High-Dimensional Linear Regression: Minimax Rates and Adaptivity"*, Sept. 2015
5. Contributed talk, Joint Statistical Meetings, Seattle, USA, *"Distance Matrix Estimation from Noisy Observation of Low Rank Position Matrix"*, Aug. 2015
6. Contributed talk, Joint Statistical Meetings, Boston, USA, *"Instrumental Variable Approach for Mediation Analysis of Count Model"*, Aug. 2014
7. Topic Contributed talk, Joint Statistical Meetings, Montreal, Canada, *" Instrumental Variable Approach for Mediation Analysis of Zero-Inflated Count Model"*, Aug. 2013
8. Poster presentation, Atlantic Causal Inference Conference, Harvard University, *"Control Function Instrumental Variable Estimation of Nonlinear Causal Effect Models"*, May. 2013

Memberships

- Institute of Mathematical Statistics
- American Statistical Association
- Royal Statistical Society
- Bernoulli Society for Mathematical Statistics and Probability
- International Chinese Statistical Association
- The Econometric Society